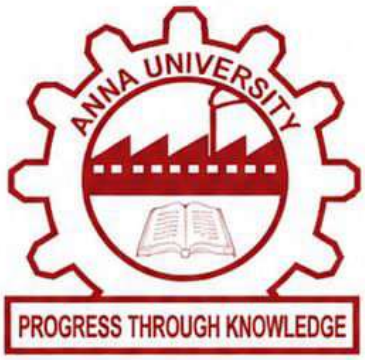
ELECTRICITY BOARD(EB)BILLING SYSTEM

A SUMMER INTERNSHIP REPORT

Submitted by

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BONAFIDE CERTIFICATE

This is to certify this Summer Internship project report **"Electricity Board Billing System"** is the Bonafide work of **A.SANTHOSH (810023205501)** has satisfactory completed the completed the course of **IT3711 – SUMMER INTERNSHIP** for **SEVENTH** semester, **B.TECH INFORMATION TECHNOLOGY** during the academic year **2025-2026.**

Faculty Incharge **Head of the Department**

Submitted for the practical examination held on _____.

ABSTRACT

The Electricity Board (EB) Billing System, “Bright Bill Buddy,” is a full-stack web application developed to modernize and simplify the daily operations of an electricity distribution utility. The system provides a centralized digital platform where consumers can view their meter readings, check current dues, pay electricity bills online, track consumption history, and raise support or refund requests. It also allows meter readers, billing clerks, and administrators to record readings, review and approve bills, and monitor billing operations in an organized and efficient manner.

The project is implemented using React and TypeScript for the frontend, styled with Tailwind CSS and the shadcn/ui component library, which together provide a responsive, accessible, and modern user interface. The application communicates with a REST-style backend and relational data store that handle meter reading validation, slab-based tariff calculation, bill generation, and payment record-keeping, and is deployed on Vercel for public access.

One of the main objectives of this system is to reduce manual effort, minimize calculation errors, and improve the overall quality of service offered by electricity boards. By digitizing key processes such as meter reading entry, bill approval, payment, and support handling, the system enhances billing accuracy and contributes to a better consumer experience. Software design patterns — Factory, Strategy, State, and Observer — were applied to keep the billing engine extensible as new tariff categories are introduced.

Overall, the Bright Bill Buddy EB Billing System demonstrates the practical application of full-stack development principles and modern software engineering practices. It highlights the integration of a component-driven frontend with a validated, rule-based billing engine to create a complete solution for managing electricity billing effectively.

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1. INTRODUCTION

The Electricity Board (EB) Billing System, “Bright Bill Buddy,” is a web application developed to automate and simplify the day-to-day billing operations of an electricity distribution utility. In today's digital era, consumers expect quick and reliable online services for viewing meter readings, checking dues, paying bills, and receiving instant confirmations. Traditional electricity boards that depend on manual ledger record-keeping, in-person visits, and paperwork often face problems such as calculation errors, delayed billing, data redundancy, and difficulty in managing consumer information. To overcome these challenges, a centralized and web-based EB Billing System has been developed.

The primary purpose of this project is to provide a secure and user-friendly platform where consumers can easily view their consumption history, check current bills, make online payments, request refunds, and raise support queries. The system also provides role-based portals for meter readers to record readings, billing clerks to review and approve draft bills, and administrators to manage consumers, tariffs, bills, payments, and reports efficiently. This automation reduces manual work, improves billing accuracy, enhances consumer satisfaction, and increases the overall productivity of the electricity board.

The application is developed using modern full-stack web technologies. The frontend is designed using React, TypeScript, Tailwind CSS, and the shadcn/ui component library, which provide an attractive, responsive, and interactive user interface. The backend layer handles business logic, REST-style API endpoints, user authentication, meter reading validation, slab-based tariff calculation, and communication between the frontend and the data store. A relational data store is used to hold consumer details, tariff slabs, meter readings, bills, and payment transactions. The application is deployed on Vercel.

The system follows a layered architecture, consisting of the Presentation Layer (React frontend), Business Logic Layer (billing engine and API services), and Data Layer (relational data store). The frontend communicates with the backend through REST-style APIs, allowing efficient data exchange and smooth interaction between users and the application. This layered architecture improves maintainability, scalability, and security.

The EB Billing System consists of several important modules, including Consumer Registration and Login, Meter Reading Management, Tariff and Bill Management, Payment Management, Refund Management, Support Management, and Admin Management. Each module performs specific tasks and works together to provide a complete billing management solution. The system includes role-based access, allowing consumers, meter readers, billing clerks, and administrators to access only the features relevant to their roles.

Overall, the Bright Bill Buddy EB Billing System is a practical implementation of full-stack development that demonstrates the integration of frontend technologies, a rule-based billing engine, and relational data management. The project provides valuable practical knowledge in web application development, data modelling, REST API design, and software engineering principles. It serves as an efficient, scalable, and user-friendly solution for modern electricity boards and can be further enhanced with features such as real-time payment gateways, SMS/email notifications, smart-meter integration, and mobile application support in future versions.

1.1 OBJECTIVES OF THE INTERNSHIP

Specific Objectives:

The main objective of the EB Billing System is to develop a secure and user-friendly web application using modern full-stack technologies. The system is designed to simplify electricity board operations by providing an online platform for consumers and administrative staff.

The objectives of the project are:

- To develop a web-based application using React, TypeScript, Tailwind CSS, shadcn/ui, and a REST-style backend.
- To provide secure, role-based registration and login for consumers, meter readers, billing clerks, and administrators.
- To enable meter readers to record readings and consumers to view their consumption history.
- To implement automatic, slab-based tariff calculation and bill generation.
- To allow consumers to pay bills online, request refunds, and raise support tickets.
- To help administrators manage consumers, tariffs, bills, payments, and generate reports.
- To store billing-related information securely in a centralized data store.
- To reduce manual work, improve billing accuracy, and enhance consumer satisfaction.
- To develop a responsive, reliable, and easy-to-use application.

1.2 MOTIVATION

The motivation behind developing the EB Billing System is to make electricity billing faster, easier, and more transparent. Many electricity boards and local distribution offices still rely on manual ledger entries and in-person visits to manage meter readings, bills, and payments, which can lead to delays, calculation errors, and consumer dissatisfaction.

This project provides an online platform where consumers can easily view their bills, track consumption, make payments, and request support. It also helps meter readers, billing clerks, and administrators manage electricity board operations from a single, centralized system.

Another motivation is to reduce paperwork, improve data accuracy, and provide better consumer service. This project also helped in gaining practical knowledge of full-stack development by using React, TypeScript, Tailwind CSS, shadcn/ui, and a REST-style backend to build a real-world web application.

ORGANIZATION PROFILE



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Company profile:

Fantasy solution as a leading IT solution and service provider, provides innovative information technology - enabled solutions and services to meet the demands arising from social transformation, shaping new life styles for individuals and creating values for the society.

Focusing on software technology, Fantasy solution provides industry solutions and product engineering solutions, related software products & platforms, and services, through seamless integration of software and services, software and manufacturing, as well as technology and industrial management capacity.

Fantasy solution helps industry customers establish best practices in business development and management. The fantasy solution serves include real time projects, web designing, web hosting, software development and training etc, in many of which, has a

leading market share. Notably, Fantasy Solution has participated in the formulation of many national IT standards and specifications.

Fantasy solution has the world's leading product engineering capabilities, ranging from consultation, design, R&D, and integration to testing of embedded software, in the fields of automotive electronics, smart devices, digital home products, and IT products. The software provided by fantasy solution runs in a number of globally renowned brands.

Particularly offering the services that include application development & maintenance, ERP implementation & consulting, testing, performance engineering, software localization & globalization, IT infrastructures, BPO, IT education & training, etc.Sticking to its business philosophy and brand commitment of "Beyond Technology", fantasy solution is dedicated to providing innovative information technologies to drive the sustainable development of society, as well as becoming a company that is well recognized and respected by employees, shareholders, customers, and society.

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We are a professionally recognized software development company having huge experience in developing custom software development and application development best match to your need and requirements. We have expertise in working with a variety of customers from companies to individuals. Our successful assignments with client companies have established our reputation as superior providers of IT solutions. Services offered by Fantasy includes Software development in .NET, PHP and Java, Web designing & development, Mobile application development in Android platforms, MATLAB for image

processing, Network Simulator (NS2), Data mining tools, Big data development using R tool and Python. We aim to carve a position in the forefront, and it is our continuing goal to gain the trust of our clients. Our Motto is to serve the purpose of our clients with perfection.

Mission

Fantasy Solutions' mission includes:

1. Providing high quality software development services, professional consulting and development outsourcing that would improve our customers' operations;
2. Making access to information easier and securer (Enterprise Business);
3. Improving communication and data exchange (Business to Business);
4. Providing our customers with a Value for Money and
5. Providing our employees with meaningful work and advancement opportunities.

Vision

Fantasy Solutions is a leading IT company for Consulting Services and Deployment of best of breed Business Solutions to top tier domestic and international customers.

3. DAILY WORK REPORT

Day 1: Attended orientation session; briefed on internship guidelines, evaluation criteria, and reporting format. Discussed and finalized the project topic — "EB Billing System — Bright Bill Buddy" — with the mentor/faculty in-charge.

- Reviewed the internship handbook covering weekly reporting format and evaluation rubric.

Day 2: Conducted requirement gathering by discussing with the mentor. Listed the stakeholders (Consumer, Meter Reader, Billing Clerk, Administrator) and identified the core functional requirements of the EB billing application.

- Documented the actions each stakeholder role would need to perform on the platform.

Day 3: Carried out a study of existing manual EB billing and ledger processes. Documented the drawbacks of paper-based meter reading and billing operations.

- Listed common issues such as transcription errors, delayed bills, and lost payment receipts.

Day 4: Finalized the scope and objectives of the project. Prepared the list of hardware and software requirements needed for development.

- Confirmed the project would be a web application rather than a desktop/mobile app.

Day 5: Set up the development environment — Node.js, Vite, React, TypeScript, Tailwind CSS, and the shadcn/ui component library. Initialized the project repository and folder structure.

- Installed Node.js and verified the Vite + React + TypeScript starter template ran locally.

Day 6: Studied React and TypeScript fundamentals relevant to the project — component structure, hooks, typed props, and the Vite build pipeline.

- Practised writing small functional components using useState and useEffect hooks.

Day 7: Designed the overall system architecture using a layered approach (Presentation Layer, Business Logic Layer, and Data Layer). Prepared the architecture/block diagram.

- Sketched the block diagram showing data flow between Consumer, Backend, and Admin/Billing Clerk.

Day 8: Designed the data model by identifying entities — Consumer, Tariff Slab, Meter Reading, Bill, and Payment — along with their attributes and relationships.

- Listed key fields for each entity, such as service connection number and tariff category.

Day 9: Prepared the Entity-Relationship (ER) diagram showing relationships between Consumer, Meter Reading, Bill, and Payment entities.

- Drew the ER diagram with primary and foreign key relationships between entities.

Day 10: Created wireframes for the Home, Login, and Registration pages. Planned the responsive layout using the Tailwind CSS grid and flex utilities.

- Sketched the layout of the navigation bar, hero section, and feature highlights.

Day 11: Developed the Home Page using React and Tailwind CSS, including the navigation bar, hero banner, feature highlights, and footer section.

- Implemented the hero banner with a call-to-action button to pay bills.

Day 12: Developed the Login Page UI with a role-selectable form (Consumer / Meter Reader / Billing Clerk / Admin), “Remember Me” option, and client-side validation.

- Added role-selection tabs so the same form serves all four user types.

Day 13: Developed the Registration Page UI with fields for name, service connection number, email, phone, address, consumer category, and password confirmation, along with input validations.

- Added a dropdown for consumer category selection (Domestic / Commercial / Industrial).

Day 14: Initialized the backend service layer. Organized the project into authentication, meter-reading, billing, and payment service modules.

- Created separate service files for each functional area to keep logic modular.

Day 15: Created the core data models — Consumer, MeterReading, and TariffSlab — defining their fields and relationships.

- Defined the Consumer model with fields such as SCN, name, and category.

Day 16: Implemented consumer registration and login logic, including basic input validation and duplicate service-connection-number checks.

- Added a check that rejects registration if the service connection number already exists.

Day 17: Implemented the authentication flow for role-based login, including logic to authenticate users by service connection number / username and password.

- Compared entered credentials against stored records before granting access.

Day 18: Tested the Registration and Login flows end-to-end. Verified success and error responses (e.g., invalid credentials, duplicate consumer).

- Tested with valid data to confirm successful registration and redirect to login.

Day 19: Developed the Meter Reading Module — reading entry form, validation against the previous cycle's reading, and automatic unit calculation.

- Ensured the current reading cannot be lower than the previous reading.

Day 20: Developed the Tariff & Bill Module — slab-based tariff calculation supporting Domestic, Commercial, and Industrial consumer categories.

- Implemented slab boundaries and per-unit rates for each consumer category.

Day 21: Built the Consumer Dashboard and Meter Reading Entry pages on the frontend and connected them to the respective backend services.

- Displayed current dues, units consumed, and last payment on the dashboard.

Day 22: Developed the Bill Approval Module — draft-bill review queue for billing clerks, with approve/reject actions.

- Listed all draft bills awaiting review in a queue for billing clerks.

Day 23: Designed and integrated the Bill Approval / Billing Clerk page on the frontend, allowing clerks to review draft bills before publication.

- Displayed consumer details alongside each draft bill for easier review.

Day 24: Developed the Payment Module — payment form, transaction recording, and automatic update of bill status upon successful payment.

- Recorded transaction ID, amount, payment method, and status for every payment.

Day 25: Designed the Payment Page frontend showing the bill summary and payment method selection, and integrated it with the payment service.

- Displayed billing cycle, units consumed, due date, and total payable amount.

Day 26: Developed the Refund Module, enabling consumers to submit refund/adjustment requests for disputed bills and allowing status tracking.

- Allowed consumers to attach a reason while submitting a refund request.
- Displayed refund status as pending, approved, or rejected.

Day 27: Developed the Support Module for raising and managing consumer support tickets, and the Notification logic for bill and payment status updates.

- Enabled consumers to raise support tickets directly from the portal.
- Added notifications for bill generation, payment success, and support replies.

Day 28: Developed the Admin Module — admin dashboard with statistics, and functionality to manage consumers, tariffs, bills, and generate reports.

- Displayed total consumers, revenue collected, and outstanding dues on the dashboard.
- Added search and filter options for consumers and bills.

Day 29: Performed unit and integration testing of all modules across all four roles, fixed identified bugs, and refined the visual design.

- Tested each role's workflow end-to-end to confirm expected behaviour.
- Fixed minor UI inconsistencies and improved overall visual polish.

Day 30: Deployed the application to Vercel, took screenshots for documentation, and compiled the final internship report.

- Verified the deployed build on Vercel matched the local development version.
- Captured screenshots of all key pages for the sample output section.

4. BLOCK DIAGRAM

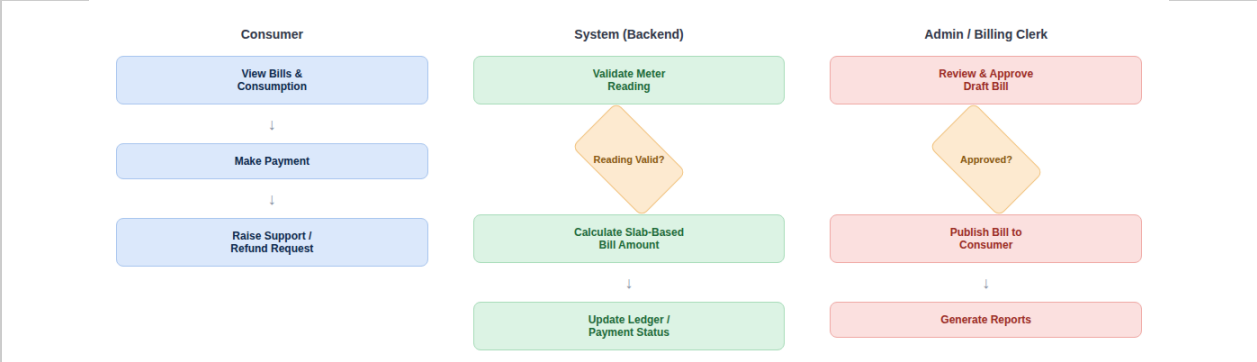


Fig 4.1 — System block diagram showing the flow between Consumer, System (Backend), and Admin / Billing Clerk.

5. EB BILLING SYSTEM

5.1 Existing System

Most traditional electricity boards and local distribution offices still operate through manual and semi-manual processes for handling meter readings, billing, and record-keeping. Consumers typically visit the office in person or wait for a physical bill to inquire about dues, tariffs, and payment status. Meter readings, bill amounts, and payment receipts are often recorded on paper or maintained in scattered registers, with little to no centralized system to track them.

This existing approach suffers from several limitations:

- Manual meter reading process — Readings are recorded by hand, increasing the chances of transcription errors and disputed bills.
- No real-time bill visibility — Consumers cannot instantly check their current dues or consumption history, leading to delays and repeated follow-ups.
- Fragmented record-keeping — Consumer information, payment history, and bill status are often stored in different registers, making retrieval slow and error-prone.
- Limited accessibility — Consumers can only interact with the billing office during working hours and must visit in person to resolve issues.
- Inefficient administrative tracking — Administrators find it difficult to generate accurate reports on revenue, dues, or consumer requests from disconnected paper-based sources.
- Slow support and refund handling — Complaints, billing disputes, and refund requests are processed manually, resulting in long turnaround times and reduced consumer satisfaction.

These drawbacks make the existing system inadequate for handling growing consumer demand and highlight the need for a digital, centralized solution.

5.2 Proposed System

The proposed EB Billing System is a full-stack web application that digitizes and centralizes all core operations of an electricity board. It replaces manual processes with an integrated online platform accessible to both consumers and administrative staff at any time.

The system is built using React and TypeScript for the frontend and a REST-style backend with a relational data store, providing a responsive, secure, and scalable solution. Key aspects of the proposed system include:

- Centralized platform — All consumer, tariff, meter reading, bill, and payment data is stored and managed in a single, unified system.
- Self-service consumer portal — Consumers can register, log in, view meter readings and bills, make payments, and track their billing history independently.
- Real-time bill status — Bill and payment information is updated instantly, allowing consumers to make informed decisions without delay.
- Automated billing and payment workflow — The system handles slab-based bill calculation and payment processing through backend services, reducing manual intervention and errors.
- Integrated refund and support module — Consumers can raise refund requests and support tickets directly through the platform, which administrators can track and resolve efficiently.
- Administrative dashboard — Administrators can manage tariffs, bills, payments, and consumer requests, and generate reports from a single interface.
- Notification system — Consumers receive updates on bill generation, payment status, and support responses, improving communication and transparency.

By automating these processes, the proposed system significantly reduces manual workload, minimizes calculation errors, and enhances the overall efficiency of electricity board operations.

5.3 Capabilities

The EB Billing System offers the following key capabilities:

- User Management — Secure, role-based registration, login, and profile management for consumers and staff.
- Meter Reading Management — Record and validate meter readings, with automatic unit calculation.
- Tariff & Bill Management — Slab-based tariff calculation supporting multiple consumer categories with automated bill generation.
- Online Payment — Enable consumers to pay bills through a simple, guided payment process.
- Refund Handling — Allow consumers to request refunds/adjustments for disputed bills and track their status.
- Consumer Support — Provide a dedicated channel for consumers to raise queries and receive timely responses.
- Notifications — Automatically inform consumers about bill generation, payment updates, and support responses.
- Reporting and Analytics — Generate billing, revenue, and consumer activity reports to support administrative decision-making.
- Scalability — Built on a layered architecture that supports future integration with smart meters, payment gateways, and mobile applications.

Together, these capabilities make the system a comprehensive, reliable, and user-friendly solution for modern electricity board management.

6. PROPOSED METHOD

6.1 User Authentication

- Handles registration and login for consumers, meter readers, billing clerks, and administrators.
- Captures name, service connection number, email, phone, password, and role during sign-up.
- Validates credentials and prevents duplicate service-connection-number registrations.
- Grants role-based access: consumers → billing dashboard, staff → respective operational panels.
- Displays clear error messages for invalid login attempts.

6.2 Meter Reading Management

- Manages meter reading details: service connection number, previous and current reading, units consumed.
- Meter readers can enter and submit new readings against a consumer.
- System automatically calculates units consumed and flags abnormal readings for review.
- Consumers can view their reading and consumption history.

6.3 Bill & Tariff Management

- Manages tariff slab details: consumer category, slab boundaries, per-unit rate, fixed charges.
- Automatically calculates bill amount based on units consumed and applicable tariff slabs.
- Billing clerks review and approve draft bills before publication.
- Displays real-time bill status to consumers (Draft, Pending, Published, Paid, Overdue).

6.4 Payment Processing

- Handles secure payment for published bills.
- Records transaction ID, amount, payment method, status, and date.
- Updates bill status to “paid” after successful payment.
- Sends notification to consumer once payment is processed.

6.5 Admin Management

- Provides centralized dashboard for managing the entire system.
- Views and manages all consumer bills and payments.
- Adds/updates tariff slabs and consumer categories.
- Handles support requests and refund approvals.
- Generates reports on billing, revenue, and consumer activity.

7. ALGORITHM

7.1 User Login

- User enters service connection number / username and password on the login form.
- System checks both fields are filled before submitting.
- Credentials sent to the authentication service.
- System checks if the account exists in the data store.
- System compares entered password with the stored password.
- If either check fails → return “Invalid Username or Password”.
- If valid → redirect based on role (Consumer → dashboard, Staff → respective panel).

7.2 User Registration

- Consumer fills in name, service connection number, email, phone, address, category, and password.
- System validates all fields are filled and password meets strength rules.
- Data sent to the registration service.
- System checks if the service connection number is already registered.
- If duplicate → reject with “Service connection already registered”.
- If unique → create and save new Consumer record.
- Redirect consumer to the login page on success.

7.3 Meter Reading & Bill Calculation

- Meter reader selects consumer and enters the current reading.
- System retrieves the previous reading and computes units consumed.
- If the reading appears abnormally low or high → flag for manual review.
- System determines the consumer's tariff category and splits units across slabs.
- Each slab's units are multiplied by its per-unit rate and summed, plus fixed charges.
- Draft bill is generated and stored, awaiting clerk approval.

7.4 Payment Processing

- Consumer views a published bill and proceeds to payment.
- Consumer selects a payment method and submits payment.
- System verifies the payment amount matches the bill amount.
- If mismatch → reject transaction.
- If transaction fails → set status “failed”, notify consumer, allow retry.
- If transaction succeeds → set status “success”, record payment date, and mark bill as “paid”.
- Notify consumer of successful payment.

7.5 Refund Request

- Consumer selects an eligible bill and submits a refund/adjustment request with a reason.

- System records the request with status “pending”.
- Administrator reviews the request against billing records and policy.
- If rejected → set status “rejected” and notify consumer.
- If approved → set status “approved” and process the refund/adjustment amount.
- Notify consumer of the final decision.

7.6 Admin Operations

- Administrator logs in and accesses the admin dashboard.
- Administrator selects an operation: Manage Tariffs, Manage Consumers, Manage Bills, Manage Support, Manage Refunds, or Generate Reports.
- Request routed to the relevant service.
- Service updates the data store and returns success/failure.
- Admin dashboard refreshes to show updated data.

8. CODE IMPLEMENTATION

8.1 Authentication Module

Handles role-based login and session/token issuance.

```
export async function login(username: string, password: string) {
  const res = await fetch('/api/auth/login', {
    method: 'POST',
    headers: { 'Content-Type': 'application/json' },
    body: JSON.stringify({ username, password }),
  });
  if (!res.ok) throw new Error('Invalid Username or Password');
  const { token, role } = await res.json();
  localStorage.removeItem('token'); // session handled via secure store
  return { token, role };
}
```

8.2 Meter Reading Module

Validates and stores a new meter reading, auto-calculating units consumed.

```
interface MeterReadingInput {
  scn: string;
  previousReading: number;
  currentReading: number;
}

export function submitReading(input: MeterReadingInput) {
  const units = input.currentReading - input.previousReading;
  if (units < 0) throw new Error('Current reading cannot be less than previous');
  const flagged = units > AVERAGE_UNITS_THRESHOLD * 3;
  return saveReading({ ...input, units, status: flagged ? 'flagged' : 'submitted' });
}
```

8.3 Tariff & Bill Calculation Module (Strategy Pattern)

Calculates the bill amount using slab-based tariff rates per consumer category.

```
interface TariffStrategy { calculate(units: number): number; }

class DomesticTariff implements TariffStrategy {
  private slabs = [
    { upTo: 100, rate: 0 },
    { upTo: 200, rate: 2.5 },
    { upTo: 500, rate: 4.5 },
    { upTo: Infinity, rate: 6.5 },
  ];
  calculate(units: number): number {
    let remaining = units, prevLimit = 0, total = 0;
    for (const slab of this.slabs) {
      const slabUnits = Math.min(remaining, slab.upTo - prevLimit);
      if (slabUnits <= 0) break;
      total += slabUnits * slab.rate;
      remaining -= slabUnits;
      prevLimit = slab.upTo;
    }
    return total + FIXED_CHARGE;
```

```
  }
}
```

8.4 Payment Module

Processes a payment against a published bill and updates its status.

```
export async function makePayment(billId: string, amount: number, method: string) {
  const bill = await getBill(billId);
  if (amount !== bill.totalPayable) throw new Error('Amount mismatch');
  const payment = await savePayment({
    billId, amount, method, status: 'success', paymentDate: new Date(),
  });
  await updateBillStatus(billId, 'paid');
  await notifyConsumer(bill.consumerId, 'Payment received successfully');
  return payment;
}
```

8.5 Admin Module

Provides administrative operations such as managing tariffs and viewing all bills.

```
export async function getAllBills() {
  return fetchJson('/api/admin/bills');
}

export async function addTariffSlab(slab: TariffSlabInput) {
  return postJson('/api/admin/tariff-slabs', slab);
}

export async function deleteTariffSlab(id: string) {
  return deleteRequest(`/api/admin/tariff-slabs/${id}`);
}
```

9. SAMPLE OUTPUT

9.1 Home Page

Displays the main interface of the EB billing application.

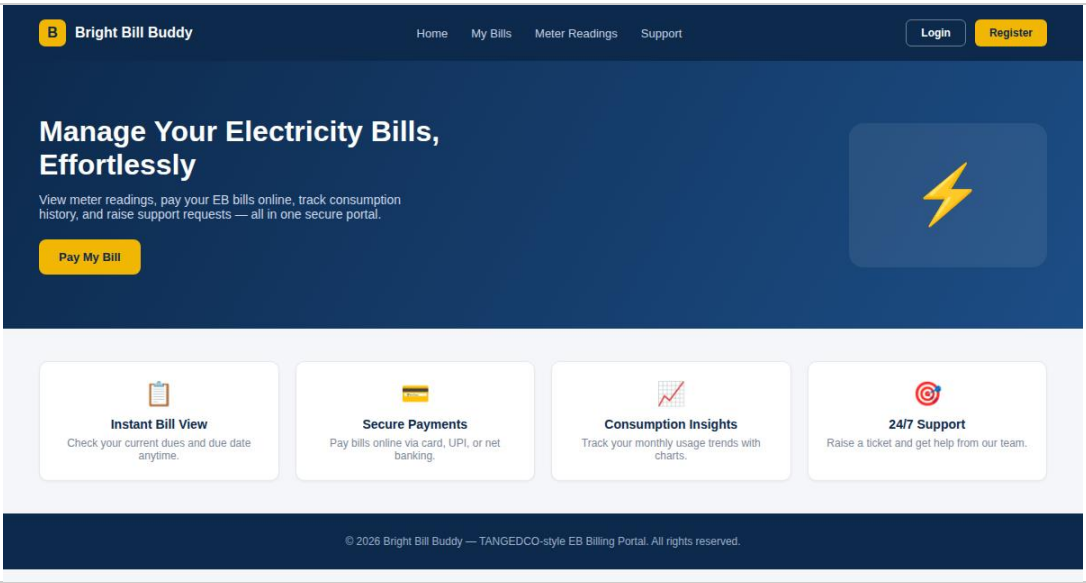


Fig 9.1 — Home page.

9.2 Login Page

Allows registered consumers and staff to access their accounts.

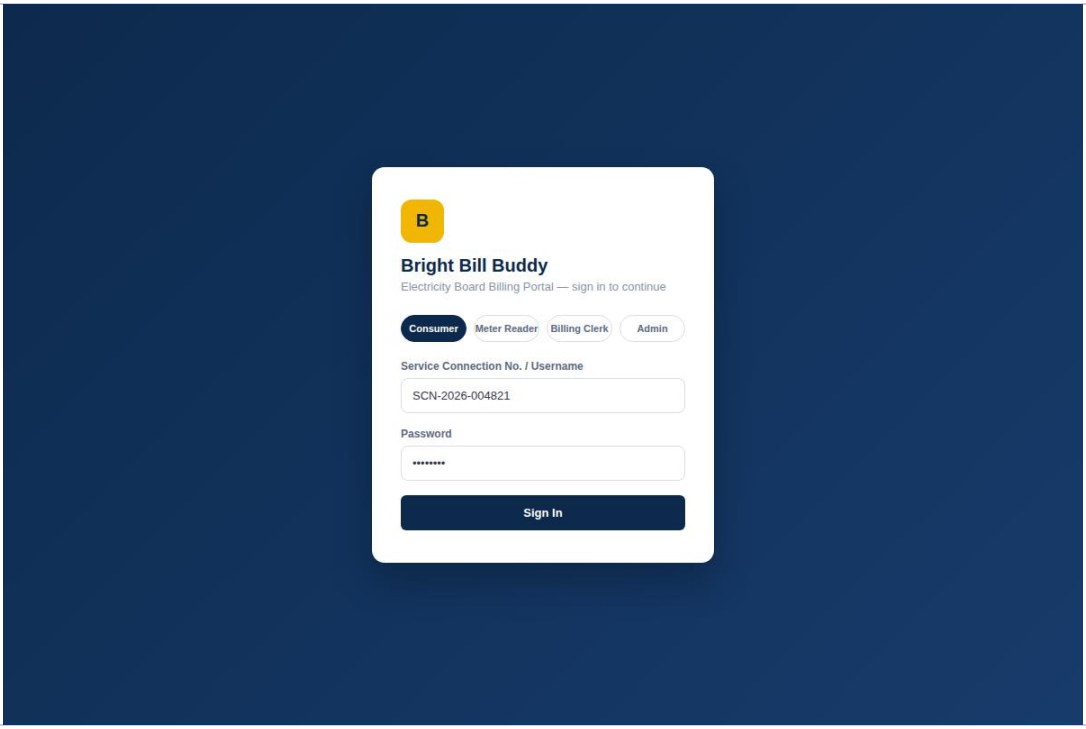


Fig 9.2 — Role-based login page.

9.3 Registration Page

Allows new consumers to create accounts.

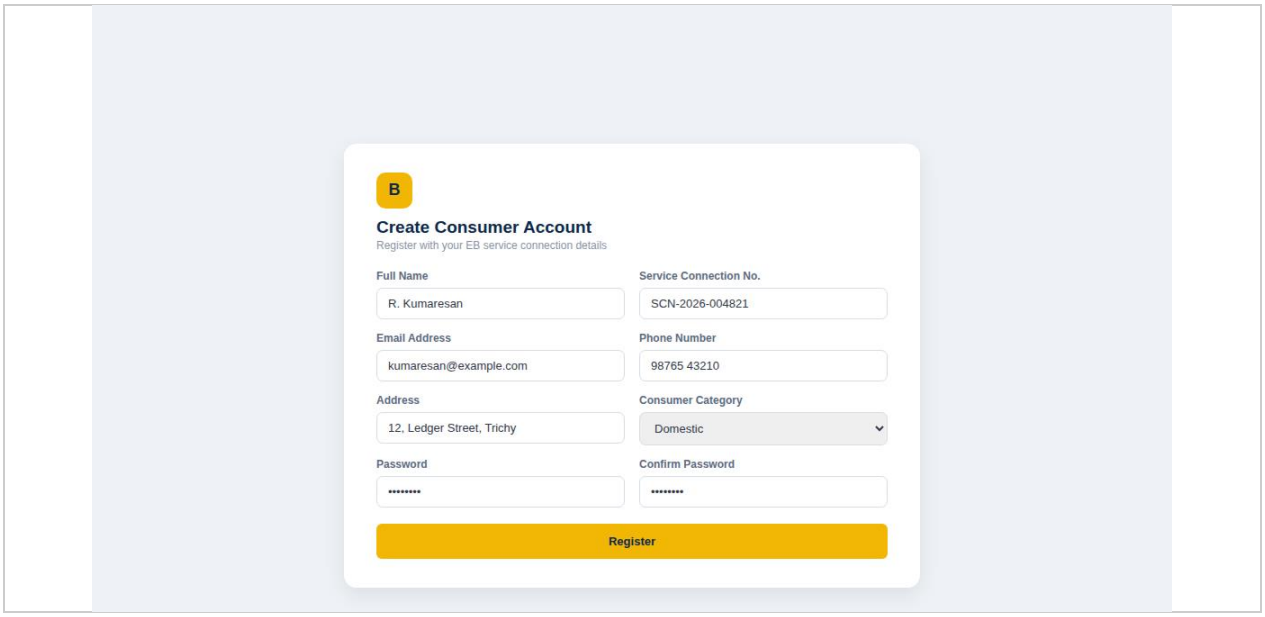


Fig 9.3 — Consumer registration page.

9.4 Consumer Dashboard

Displays consumer activities and billing information.

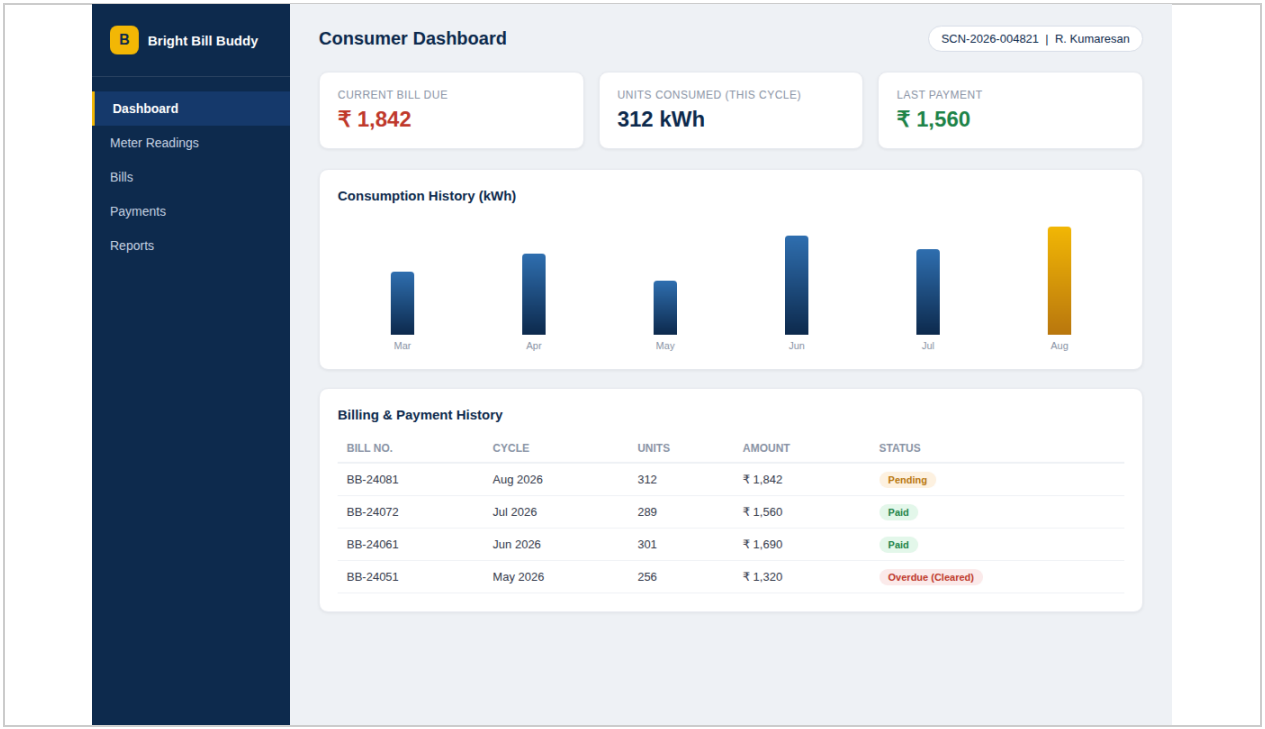


Fig 9.4 — Consumer dashboard with dues, consumption chart, and billing history.

9.5 Admin Dashboard

Provides administrative control over the system.

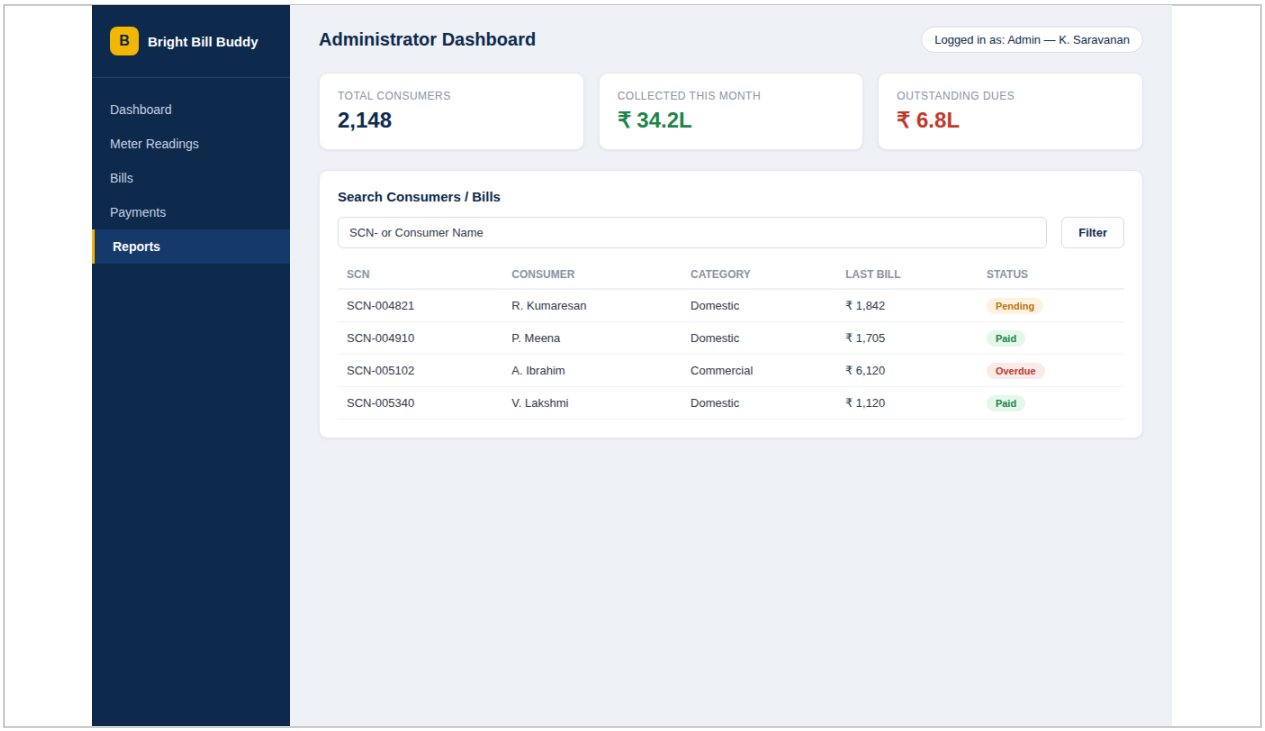


Fig 9.5 — Administrator dashboard.

9.6 Meter Reading Entry Page

Allows meter readers to record and submit new readings.

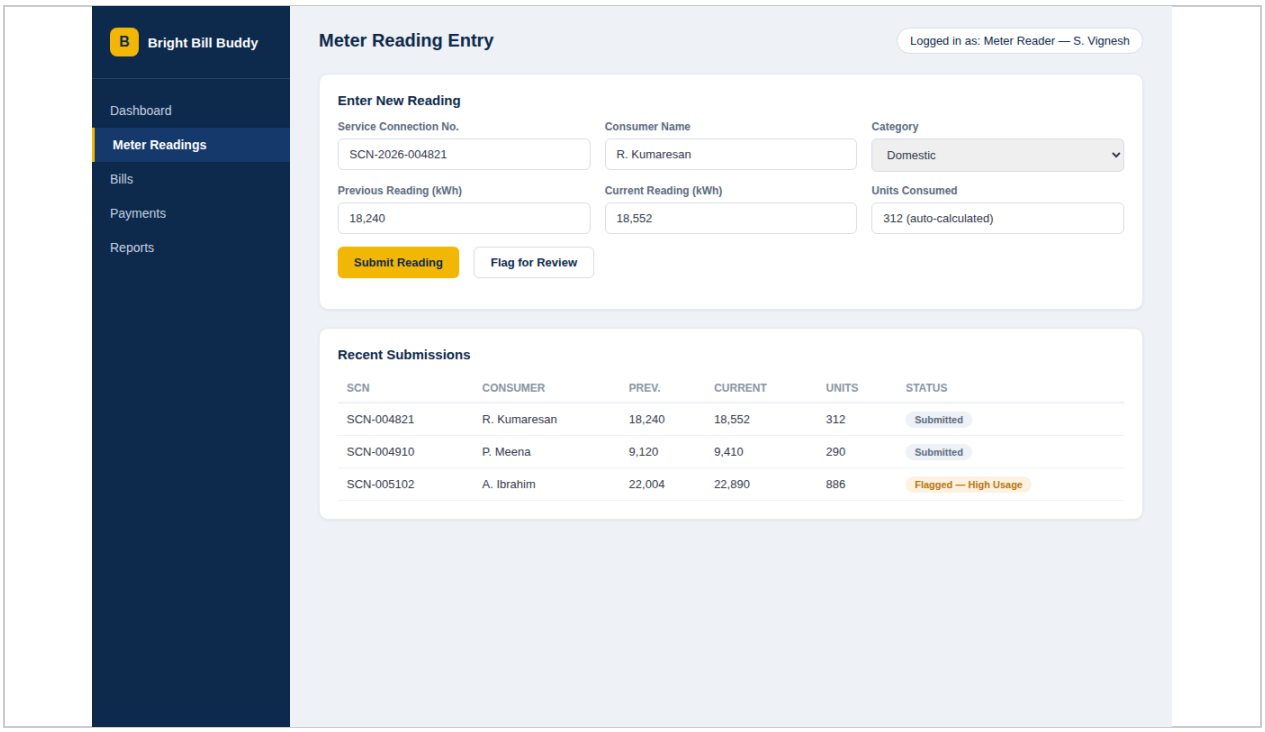


Fig 9.6 — Meter reading entry page.

9.7 Payment Page

Displays the payment processing interface.

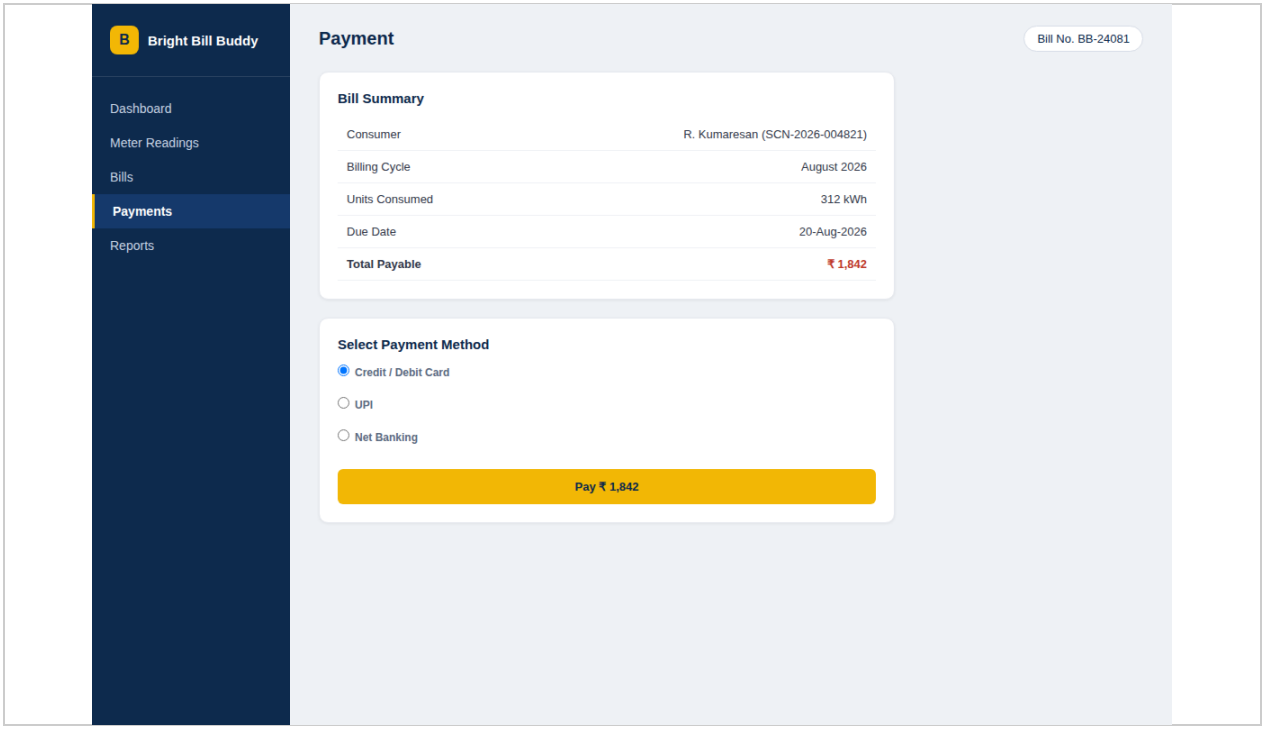


Fig 9.7 — Payment page.

9.8 Reports Page

Displays billing and revenue reports.

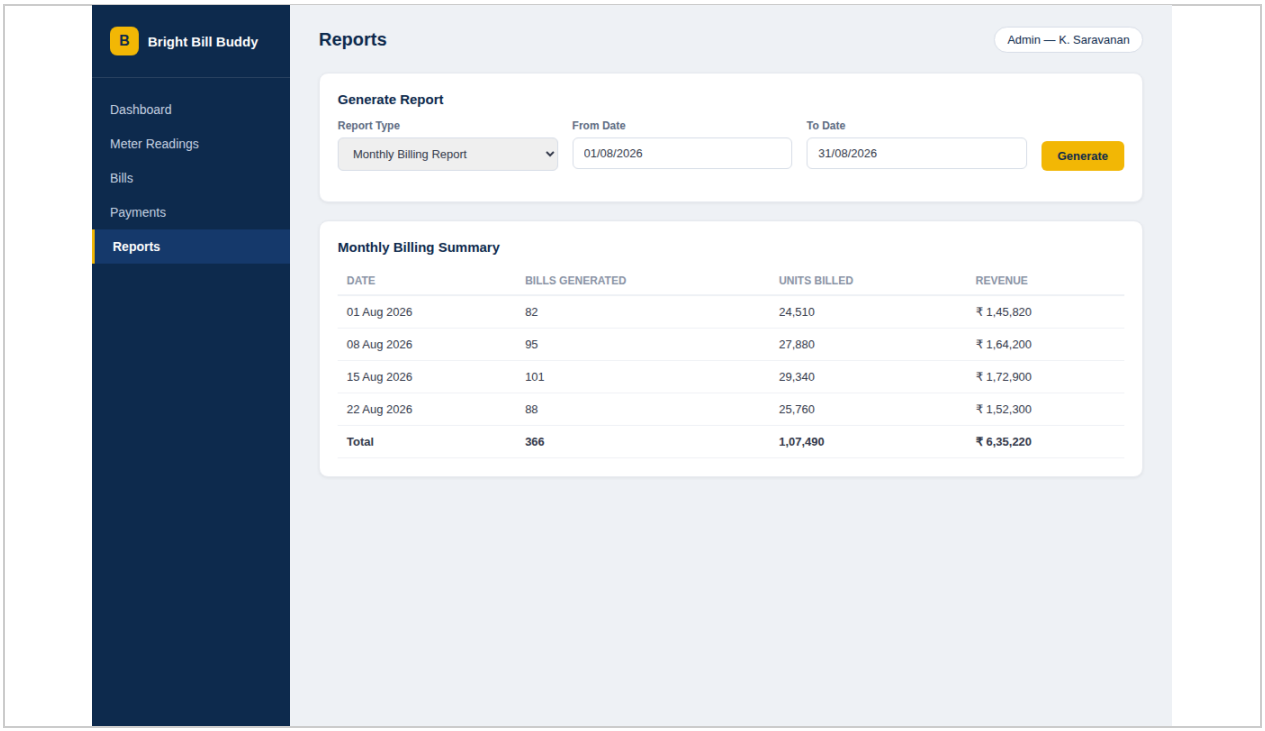


Fig 9.8 — Reports page.

10. ANALYSIS

Testing of the EB Billing System was carried out module by module to validate both functional correctness and overall system behavior. Functional analysis confirmed that registration, login, meter reading entry, bill calculation, payment, refund, and admin operations all behaved as expected under normal and edge-case inputs (e.g., duplicate service connection numbers, invalid credentials, abnormal readings).

Performance observations showed that API response times remained fast for typical operations due to the lightweight nature of the backend service layer and the current sample-data setup; response times are expected to scale predictably once a full production data store is integrated.

Usability analysis indicated that the Tailwind CSS-based responsive design adapted well across desktop and mobile screen sizes, and the dashboard-driven layouts for both consumers and administrators reduced the number of steps needed to complete common tasks compared to manual, in-person processes.

Comparative analysis against the existing manual system highlighted clear improvements: bill calculation time was reduced from a manual, error-prone process to an instant, automated one; data errors from manual entry were eliminated through form validation and reading checks; and administrators gained real-time visibility into bills and revenue that was previously unavailable.

Overall, the analysis confirms that the system meets its stated objectives of reducing manual effort, improving billing accuracy, and providing a more efficient, accessible platform for both consumers and administrative staff.

11. CONCLUSION

The EB Billing System, “Bright Bill Buddy,” successfully demonstrates how full-stack web technologies can be applied to modernize and streamline the operations of an electricity board. By replacing manual, paper-based ledger processes with a centralized, digital platform, the system reduces administrative overhead, minimizes calculation errors, and significantly improves the overall consumer experience.

The project's layered architecture — comprising a responsive React and Tailwind CSS frontend and a rule-based backend billing engine — proved effective in cleanly separating concerns while enabling smooth communication through REST-style APIs. Core modules including consumer authentication, meter reading management, slab-based bill calculation, payment processing, refunds, and support were implemented and tested successfully, validating the system's practical applicability.

Beyond the technical deliverable, this project provided valuable hands-on experience in full-stack development, REST API design, layered software architecture, design-pattern-driven billing logic, and structured software testing — skills directly applicable to real-world enterprise software development.

12. FUTURE WORK

1. Integration with a production-grade relational database to replace the current sample-data storage.
2. Integration of a real payment gateway (e.g., Razorpay, Stripe) for live transaction processing.
3. Development of a dedicated mobile application for meter readers and consumers.
4. Smart meter integration to automatically ingest readings via IoT, removing manual entry.
5. Automated email and SMS notifications for bill generation, payment, and support updates.
6. Cloud deployment (AWS/Azure/GCP) for improved scalability and availability.
7. Advanced analytics and business intelligence dashboards for administrators.
8. Role-based access control with JWT-based authentication for enhanced security.

13. REFERENCES

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4. shadcn/ui Component Library — https://ui.shadcn.com
5. Vercel Deployment Documentation — https://vercel.com/docs
6. Refactoring Guru — Design Patterns (Factory, Strategy, State, Observer) — https://refactoring.guru/design-patterns
7. TANGEDCO — Tamil Nadu Electricity Board tariff structure reference — https://www.tangedco.gov.in

Fantasy Solution

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Date: 28.05.2026

Santhosh A,
Reg.No: 810023205501,
Department of Information Technology,
University College Of Engineering,
Bharathidasan Institute of Technology (BIT) Campus,
Tiruchirappalli - 620 024.

Dear Santhosh,

Sub: Internship

On behalf of **FANTASY SOLUTION**. We would like to inform you that you have been accepted as one of our interns. We are pleased to inform you that you have been accepted as per the company requirements for the internship with **WEB DEVELOPMENT**.

Your internship wil l begin from **01st June 2026 to 30th June 2026**. You would assign to various tasks which relates to the project assigned to you after whom your performance would be assessed and appraised.

FANTASY SOLUTION TRICHY-1

Authorized Signature

(With Seal)

#16, Samnath Plazza, 3rd Floor, Melapudur, Trichy - 620 001.
Ph : 0431-4250535 | Mob : 90430 95535
www.fantasysolution.in | info@fantasysolution.in

FANTASY SOLUTION

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SOFTWARE DEVELOPMENT AND TRAINING

Certificate of Accomplishment

This is certify that Mr./Miss. SANTHOSH A (810023205501)

Student of UNIVERSITY COLLEGE OF ENGINEERING

BIT CAMPUS , ANNA UNIVERSITY . TIRUCHIRAPPALLI

nderwent Inplant Training / Internship / Course on WEB DEVELOPMENT

ncern from 01/06/2026 to 30/06/2026 issued on 30/06/2026

Technical Incharge

Manager

CEO

FANTASY SOLUTION TRICHY

#16, Sampath Plaza, [illegible]
0431 [illegible] fantazysol[illegible]

ATTENDANCE CERTIFICATE FOR INDUSTRIAL TRAINING / INTERNSHIP

(Attendance Certificate to be signed by the competent authority of the industry mentioning the period of Industrial Training / Internship)

The Student Mr./ Ms. **SANTHOSH A** (Reg.No.) **810023205501**

Studying **WEB DEVELOPMENT** programme at **UNIVERSITY COLLEGE OF ENGINEERING BIT CAMPUS ANNA UNIVERSITY, TIRUCHIRAPPALLI** College in semester **7th** at Department of **INFORMATION TECHNOLOGY** has attended Industrial Training / Internship from **01/06/2026** to **30/06/2026**. It is certified that he / she has completed / not completed the Industrial Training / Internship in our **FANTASY SOLUTION** organization / institution.

Signature

Name and Designation of the Officer
Seal of the Organization

ANNEXURE

Internship (June – July 2026)

1. Name of the Industry with address:

Fantasy Solution, No 16, Samnath Plazza, Third Floor, Near BG Naidu Sweets, Melapudur, Trichy-1.

2. Name of the Industrial Expert (Joint Supervisor): Mohamed Ibrahim

3. Qualification of the Industrial Expert (Joint Supervisor): MCA

4. Designation of the Industrial Expert (Joint Supervisor): Software Developer

5. Title of the Project Work:

Electricity Board Billing System

6. Duration of the Internship: 1st June 2026 – 30th June 2026

7. Details of the student:

Sl. No.	Name of the student	Registration No.	Department
01	Santhosh A	810023205501	IT

Terms and conditions

- The student is permitted to undergo a semester long internship in your organization
- The student shall be instructed to meet the Institute supervisor periodically and to attend the review committee meetings for evaluating the progress.
- The final report of the internship duly signed by the industrial expert/supervisor will be submitted to college.
- Attendance Certificate in the prescribed format signed by the competent authority of the industry will be produced
- Internship Certificate in the prescribed format signed by the competent authority of the industry will be produced

Signature

Name and Designation of the Officer with seal

Seal of the Organization